



Predicting Burnout in College Students: A Machine Learning Approach Using Decision Tree and Psychometric Data

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Author's contribution

The sole author designed, analysed, interpreted and prepared the manuscript.

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Abstract

Aims: This study aimed to model and predict academic burnout among college students using psychological and academic variables, and to identify the most influential factors contributing to burnout risk.

Study Design: This research employed a quantitative, predictive research design utilizing decision tree analysis to classify students into burnout and non-burnout categories.

Place and Duration of Study: The study was conducted at a private higher education institution in Misamis Occidental, Philippines, from January to March 2025.

Methodology: A total of 291 college students participated through convenience sampling using an online survey. The instruments used included the Maslach Burnout Inventory (MBI), a tool designed to measure emotional exhaustion, cynicism, and academic efficacy along with the DASS-21 for stress, anxiety, and depression, the Brief COPE inventory, and a sleep quality measure. Descriptive statistics were used to profile respondents' demographics and psychological traits. A Decision Tree model was applied to classify students as "burnout" (MBI score ≥ 3.0) or "not burnout," and feature importance was analyzed to determine key

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predictors. Model performance was assessed using Recall, F1-score, and the Receiver Operating Characteristic–Area Under the Curve (ROC-AUC), which evaluates the model’s ability to distinguish between burnout and non-burnout cases.

Results: Out of 291 respondents, 139 (47.8%) were classified as experiencing burnout, while 152 (52.2%) were not. Stress score (29.5%), coping strategies (26.4%), and sleep duration (15.4%) emerged as the top predictors. The model achieved a recall of 51.6%, F1 score of 59.3%, and ROC-AUC of 66.9%, indicating moderate predictive power. Descriptive analysis revealed that most respondents were male and reported moderate stress and good sleep quality, which may contribute to the lower burnout rates observed.

Conclusion: Stress, coping strategies, and sleep behaviors are the most influential factors in predicting student burnout. It is recommended that future studies consider a female-majority sample and use more advanced machine learning models such as Random Forest to improve prediction accuracy and understanding of burnout patterns.

Keywords: Academic burnout; decision tree; college students; stress, coping strategies; sleep quality; psychometric assessment; predicting modeling.

1 Introduction

Academic burnout is increasingly prevalent in higher education, especially at private universities, where students often face a combination of elevated academic demands, financial pressure, and restricted access to mental health supports or effective coping strategies. In contrast to transient stress, burnout is a chronic psychological disorder characterized by emotional tiredness, detachment or cynicism toward academic responsibilities, and a reduced feeling of one's academic efficacy (Dyrbye et al., 2006; Salamonson et al., 2016). These symptoms adversely affect a student’s psychological health and hinder academic achievement, leading to absenteeism, disengagement, and, in extreme instances, institutional dropout. The atmosphere in private universities frequently exacerbates these demands owing to the elevated cost of education and the competitive academic culture, which may not consistently provide sufficient support networks. Students may be balancing demanding academics with financial obligations and personal expectations, all while missing substantial institutional support for mental health and well-being (Campos et al., 2012). This amalgamation of stressors elevates their likelihood of experiencing burnout.

Current data robustly correlates academic burnout with numerous psychological and behavioral risk factors. Elevated stress levels, inadequate sleep quality, and the employment of maladaptive coping mechanisms—such as avoidance, self-blame, or disengagement—often serve as primary factors in the manifestation of burnout symptoms (Beiter et al., 2015; Yang & Farn, 2005; Galambos et al., 2009). These behavioral and psychological states never exist alone and frequently intersect with symptoms of anxiety and depression, exacerbating the difficulty of accurately detecting and resolving burnout. Researchers have utilized well-validated psychometric instruments to comprehend and quantify these intricate psychological dynamics. The Maslach Burnout Inventory – Student Survey (MBI-SS) evaluates essential elements of burnout, including weariness, cynicism, and diminished efficacy (Schaufeli et al., 2002). The Depression, Anxiety, and Stress Scale (DASS-21) offers insight into general emotional distress, whereas the Brief COPE Inventory examines the coping strategies employed by students to address stress, encompassing both adaptive behaviors such as planning and problem-solving and maladaptive behaviors like denial and behavioral disengagement (Matud, 2004). Collectively, these instruments provide an extensive overview of student mental health and resilience.

Notwithstanding the efficacy of these instruments, the interpretation of data derived from them generally depends on conventional statistical techniques. Methods such as correlation, linear regression, and fundamental comparative analysis, although useful, frequently inadequately capture the complex, nonlinear relationships among psychological variables. These methodologies may conceal the nuanced interdependencies and risk patterns that more sophisticated analytical tools can uncover (Zhou et al., 2020). Consequently, educational institutions may overlook early indicators of burnout or miss critical opportunities for targeted assistance. This signifies a considerable constraint in the utilization of psychological data to enhance student progress and mental well-being.

This paper presents an integrative strategy that amalgamates current psychological evaluation instruments with machine learning techniques, particularly Decision Tree algorithms, to address these shortcomings. Decision Trees are a type of supervised learning that delineates rule-based associations between input variables and

outcomes in a manner that is both predictive and easily interpretable (Kotsiantis, 2012). In contrast to more obscure models like neural networks or ensemble methods, Decision Trees offer transparent, visual representations that assist stakeholders—academic advisors, counselors, and administrators—in comprehending how various combinations of factors, such as elevated stress and insufficient sleep, lead to burnout. This transparency renders them especially beneficial in practical educational environments, where actionable insights must be readily comprehensible to practitioners lacking technical proficiency in data science.

While prior research in educational data mining has illustrated the effectiveness of Decision Tree algorithms in forecasting academic outcomes like student performance, retention, and engagement, there is a paucity of studies that have utilized this methodology specifically for predicting and modeling academic burnout through psychometric data. This work seeks to address that gap by developing a prediction model specifically designed for the burnout environment, utilizing a sample of undergraduate students from a private university. Data will be gathered via the MBI-SS, DASS-21, and the Brief COPE Inventory, in conjunction with a researcher-created sleep quality scale that encompasses both quantitative and qualitative aspects of sleep behavior. This dataset will serve as a comprehensive basis for identifying principal predictors of burnout and understanding the mechanisms via which burnout arises (Beiter et al., 2015).

This research aims to quantify the prevalence and severity of academic burnout within this student demographic and to identify the key psychological and behavioral factors that contribute to it. This work employs Decision Tree modeling on multidimensional data to create a tool for institutions to identify students at danger of burnout prior to further decline in academic performance and mental health. The decision tree model will provide a visual and evidence-based diagnostic framework that facilitates early diagnosis and focused intervention, allowing academic institutions to adopt a more proactive and individualized approach to student mental health (Lund et al., 2010; Zhou et al., 2020).

This research enhances both theoretical comprehension and practical implementation. It addresses a timely and underexplored intersection between psychological evaluation and machine learning, offering tools that can be used by educational institutions to preemptively detect burnout. The study contributes to the broader field of educational data science by demonstrating the practical value of interpretable machine learning models in real-world mental health interventions. Moreover, it supports evidence-based policymaking for student support programs by identifying key predictors of burnout through transparent, accessible tools. As such, it holds relevance for both researchers and practitioners committed to improving student well-being and academic resilience.

It theoretically improves understanding of how intricate connections among stress, coping mechanisms, emotional distress, and sleep patterns lead to academic burnout. It provides a predictive and interpretable instrument that may be integrated into institutional wellness and advising systems. The study integrates psychological assessment with data-driven decision-making, addressing a significant gap in student support and providing a framework for higher education institutions to more effectively tackle the mental health difficulties of contemporary students.

2 Methodology

2.1 Research design

This study modelled and predicted academic burnout among college students using a quantitative predictive research methodology. It sought to create a predictive model based on significant behavioral and psychological factors such as stress, anxiety, depression, coping strategies, and sleep quality using the machine learning technique of Decision Tree Classification. A standardized questionnaire consisting of well-known instruments such as the Maslach Burnout Inventory – Student Survey (MBI-SS) (Schaufeli et al., 2002), the Depression, Anxiety, and Stress Scale (DASS-21) (Lovibond & Lovibond, 1995), and the Brief COPE Inventory (Carver, 1997) was combined with a Sleep Quality Scale created by the researcher. JAMOMI pilot testing was used to assess the Sleep Quality Scale's internal consistency and reliability; Cronbach's alpha value of 0.70 or more was considered satisfactory. The final data processing, model training, and model evaluation were all done in Python. Model performance was evaluated using **Recall**, **F1 Score**, and the **Receiver Operating Characteristic–Area Under the Curve (ROC-AUC)** to assess its sensitivity, predictive balance, and ability to distinguish between burnout and non-burnout classes. To guarantee the Decision Tree's usefulness in educational contexts, model interpretability was also stressed.

2.2 Research respondents

The respondents of this study were 291 college students who were enrolled in a private higher education institution. Participants were chosen using a convenience sample technique, which made it possible for the researcher to collect data fast and effectively. Given the limited time and resources available during the study's conduct, convenience sampling was selected for its accessibility and practicality.

The researcher used an online Google Form survey to collect data, enabling participants to respond at their convenience and from a distance. Social media and government communication channels were used to spread the survey link. There was no requirement to participate, and pupils had enough time to fill out the form on their own. Prior to analysis, the researcher verified that all submissions were valid and complete and kept an eye on response rates.

2.2.1 Demographic and academic characteristics of respondents

The final sample consisted of 291 college students from a private higher education institution. Of the total, 198 (68.04%) were male and 93 (31.96%) were female. In terms of academic year, 98 (33.68%) were first-year students, 77 (26.46%) were in their second year, 106 (36.43%) were in their third year, and 10 (3.44%) were in their fourth year. Nearly half of the respondents (47.77%) were enrolled in the College of Business Administration (CBA), followed by the College of Teacher Education (CTE) with 14.78%, and the College of Computer Studies, Engineering, and Architecture (CCSEA) with 13.06%.

The mean age of participants was 20.27 years ($SD = 2.03$). On average, students studied for 3.23 hours per day ($SD = 1.96$), were enrolled in 25.89 academic units ($SD = 5.13$), and reported an average of 5.94 hours of sleep per night ($SD = 1.30$). These figures provide context for evaluating psychological variables related to burnout, such as stress, coping strategies, and sleep quality.

2.3 Instrument of the study

Four tools were used in this study: the Maslach Burnout Inventory – Student Survey (MBI-SS), the Depression, Anxiety, and Stress Scale (DASS-21), the Brief COPE Inventory, and a Sleep Quality Scale created by the researcher. The Sleep Quality Scale was created especially for this study based on the body of research on sleep evaluation, whereas the MBI-SS, DASS-21, and Brief COPE are widely validated instruments taken from other studies. The reliability of the researcher-created Sleep Quality Scale was evaluated using Cronbach's Alpha in the JAMOV program (version 2.6.23). With an alpha value above 0.70, the instrument demonstrated adequate internal consistency. As such, the tool is regarded as reliable for assessing college students' sleep quality.

2.3.1 Maslach Burnout Inventory – Student Survey (MBI-SS)

The following variables are used in this tool to assess students' experiences of academic burnout: academic efficacy, cynicism, and emotional exhaustion. A 6-point Likert scale (0 = Never to 5 = Every Day) is used to evaluate the 15 items on the test. Burnout experiences are categorized from No Experience to Very High using pre-established mean ranges. This tool consists of three subscales: Emotional Exhaustion (feeling drained by academic work), Cynicism (detachment from studies), and Academic Efficacy (sense of competence and achievement).

Table 1. Descriptive category for academic burnout (MBI-SS)

Scale	Mean Range	Descriptive Rating
5	4.34 – 5.00	Very High
4	3.67 – 4.33	High
3	3.00 – 3.66	Moderate
2	2.34 – 2.99	Low
1	1.67 – 2.33	Very Low
0	0.00 – 1.66	No Experience

2.3.2 Depression, Anxiety, and Stress scale (DASS-21)

This instrument assesses college students' emotional states of stress, anxiety, and sadness. It has 21 items with the following choices, each assessed on a 4-point Likert scale:

4 = Applied to me very much or most of the time, 3 = Applied to me to a large degree, 2 = Applied to me to some degree, and 1 = Did not apply to me at all.

The results are categorized from Normal to Extremely Severe emotional states using pre-established mean ranges. The ranges displayed here are used to interpret each participant's mean score. The scale includes three subscales measuring symptoms of Depression (e.g., hopelessness), Anxiety (e.g., physiological arousal), and Stress (e.g., difficulty relaxing or irritability).

Table 2. Descriptive Category for Depression, Anxiety, and Stress (DASS-21)

Scale	Mean Range	Descriptive Rating
4	3.25 – 4.00	Extremely Severe
3	2.50 – 3.24	Severe
2	1.75 – 2.49	Moderate
1	1.00 – 1.74	Normal

2.3.3 Brief COPE inventory

This instrument assesses students' coping processes and methods for handling both personal and academic challenges. It has 28 items with the following choices, each assessed on a 4-point Likert scale: One indicates that I haven't been doing this at all, two indicates that I have been doing this a little bit, three indicates that I have been doing this a medium amount, and four indicates that I have been doing this extensively. To categorize the frequency of coping techniques from Rarely to Very Frequent, the results are examined using predefined mean ranges. The ranges displayed here are used to interpret each participant's mean score. The inventory includes subscales for adaptive coping strategies (e.g., planning, active coping, use of emotional support) and maladaptive coping (e.g., denial, self-blame, behavioral disengagement).

Table 3. Descriptive Category for Coping Strategies (Brief COPE)

Scale	Mean Range	Descriptive Rating
4	3.25 – 4.00	Extremely Severe
3	2.50 – 3.24	Severe
2	1.75 – 2.49	Moderate
1	1.00 – 1.74	Normal

2.3.4 Researcher-made sleep quality scale

This tool assesses students' sleep habits and general quality of sleep, paying particular attention to factors including sleep onset, nightly awakenings, daytime weariness, and sleep satisfaction. Ten items with the following options are assessed on a 4-point Likert scale:

1 denotes never, 2 sometimes, 3 often, and 4 always. The results are categorized from Very Poor to Very Good sleep quality using pre-established mean ranges. The ranges displayed here are used to interpret each participant's mean score. The scale covers aspects such as sleep latency, sleep interruptions, daytime fatigue, and overall satisfaction with sleep.

Table 4. Descriptive category for sleep quality (Researcher-made scale)

Scale	Mean Range	Descriptive Rating
4	3.25 – 4.00	Very Good Sleep Quality
3	2.50 – 3.24	Good Sleep Quality
2	1.75 – 2.49	Poor Sleep Quality
1	1.00 – 1.74	Very Poor Sleep Quality

2.4 Data gathering procedure

Four validated tools were used to collect data: the Brief COPE Inventory, the Depression, Anxiety, and Stress Scale (DASS-21), the Maslach Burnout Inventory – Student Survey (MBI-SS), and a Sleep Quality Scale created by the researcher. To ensure ease of use and accessibility, all tools were transformed into Google Forms. Participants received the link and were given enough time to fill out the forms on their own. Before starting data analysis, the researcher kept an eye on the responses, reminded participants to finish any questions that remained, and made sure that all submissions were complete.

2.5 Statistical treatment of data

The demographic profiles and psychometric scores of the participants were summarized using descriptive statistics, such as means and frequency distributions, derived from the Maslach Burnout Inventory – Student Survey (MBI-SS), the Brief COPE Inventory, the Depression, Anxiety, and Stress Scale (DASS-21), and the Sleep Quality Scale.

Using the psychometric characteristics as indicators of academic burnout state, a Decision Tree Classification algorithm was used to create the prediction model. Students were divided into "Burnout" and "No Burnout" groups according to the MBI-SS's defined scoring parameters, which were used to determine the burnout outcome variable.

The following statistical metrics were used to assess the Decision Tree model's predictive performance: Recall (Sensitivity) – the percentage of actual burnout cases that the model correctly identified; F1 Score – which balances the trade-off between precision and recall by taking their harmonic mean; and Receiver Operating Characteristic–Area Under the Curve (ROC-AUC) – which measures the model's ability to distinguish between burnout and non-burnout classifications across various thresholds. Additionally, feature importance scores were computed to determine which psychometric and academic variables contributed most significantly to the model's predictions. Variables such as stress levels, coping strategies, and sleep patterns emerged as key predictors of academic burnout. A 5-fold cross-validation procedure was applied to enhance the model's reliability. Emphasis was placed on model interpretability to facilitate practical application in educational and psychological counseling settings.

3. Results and Discussion

This section delineates the conclusions of the study on modeling student academic burnout. Descriptive data are presented to illustrate overarching tendencies among respondents, encompassing mean scores for study hours, stress and depression levels, coping mechanisms, and sleep quality. A decision tree analysis was conducted to ascertain the most critical determinants of academic burnout. The feature importance output indicates the relative contribution of each variable in assessing burnout risk among students.

The decision tree model's predictive ability was assessed by classification accuracy, precision, recall, and the F1-score, with the identification of key predictors. These metrics provide a thorough evaluation of the model's efficacy in identifying academic burnout cases and reducing misclassification. The data are analyzed about their significance for early identification and intervention tactics.

3.1 Descriptive analysis of psychological scales

Sleep quality, burnout symptoms, stress, sadness, anxiety, and coping mechanisms are critical psychological indicators that might affect a student's academic well-being. This section delineates the descriptive data of the psychological scales employed in the study, encompassing the Maslach Burnout Scale, DASS-21 (Stress, Anxiety, Depression), and the Brief COPE inventory. These methods offer a thorough assessment of the respondents' psychological and emotional conditions, essential for detecting probable burnout. Comprehending these principles provides insight into students' management of academic pressures and the potential early indicators of burnout.

Table 5 indicates that the respondents exhibited a low degree of burnout ($M = 2.92$), satisfactory sleep quality ($M = 2.63$), and moderate degrees of stress ($M = 2.19$), anxiety ($M = 2.11$), and sadness ($M = 1.92$). The results suggest that the subjects are not presently undergoing significant psychological anguish. The mean sleep

duration per night ($M = 2.60$) categorizes as "severe" sleep deprivation, indicating a potential long-term harm to well-being if maintained. The data indicated that most responders were male. Research indicates that female students generally exhibit elevated emotional reactivity, tension, and rumination relative to their male counterparts (Matud, 2004; Nolen-Hoeksema, 2012), thereby elucidating the diminished stress and burnout scores recorded in our sample. Additionally, research by Lund et al. (2010) demonstrated that high sleep quality markedly alleviates symptoms of anxiety and sadness in college students, reinforcing the idea that the generally favorable sleep quality observed in this study may enhance their moderate emotional ratings. These findings emphasize the importance of sleep and stress management techniques in averting academic burnout.

Table 5. Descriptive Statistics of Psychological Scales

Variable	Mean	Interpretation
Maslach Burnout Scale	2.92	Low
Sleep Quality	2.63	Good Sleep Quality
DASS-22 Stress	2.19	Moderate
DASS -22 Anxiety	2.11	Moderate
DASS-22 Depression	1.92	Moderate
Sleep Hours per Night	2.60	Severe

3.2 Decision tree analysis

This section demonstrates the utilization of the Decision Tree algorithm to categorize students according to their susceptibility to academic burnout. The investigation utilized psychometric and academic data to ascertain whether students are predisposed to burnout and to find the elements that most significantly contributed to this classification. The subsequent subsections delineate (1) the model's categorization outcomes, (2) the principal features utilized for prediction, and (3) the assessment metrics that characterize the model's success.

3.2.1 Burnout classification output

This section delineates the outcomes of the Decision Tree model in categorizing pupils into burnout classifications based on psychometric and academic data. The model was trained utilizing the Maslach Burnout Inventory (MBI), with a mean score of 3.0 or higher signifying academic burnout. The objective was to identify at-risk pupils by analyzing patterns from psychological variables, including stress, anxiety, coping mechanisms, and sleep behavior.

According to the model, 139 students (47.8%) were identified as experiencing academic burnout, whereas 152 (52.2%) were not. Fig. 1 illustrates this distribution in a pie chart for enhanced visualization. Although nearly half of the respondents were at risk, a marginally greater number did not satisfy the criteria for burnout.

Predicted Burnout Classification (Decision Tree Mode

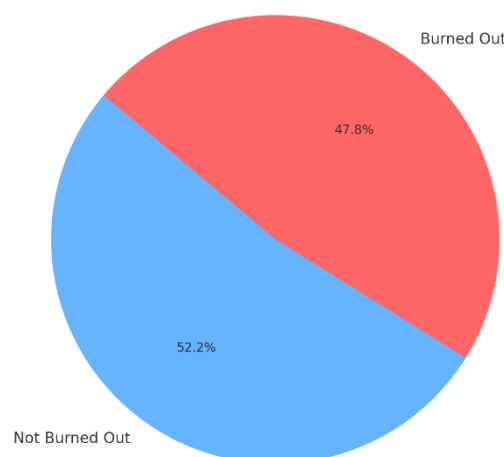


Fig. 1. Predicted burnout classification

A contributing aspect to the reduced burnout rate may be the gender composition of the sample, as most responders were male. Research indicates that female students generally exhibit elevated levels of stress, emotional weariness, and rumination—elements associated with burnout—relative to their male counterparts (Matud, 2004; Nolen-Hoeksema, 2012). This may elucidate the comparatively diminished average stress and burnout levels noted in this study. Nonetheless, the reality that nearly fifty percent of the sample is at risk underscores the persistent psychological strain in higher education and the necessity for targeted mental health support.

3.2.2 Feature importance

While most students were deemed not to be experiencing academic burnout, a substantial portion still exhibited indicators of being at risk. The feature importance output of the Decision Tree model was evaluated to identify the psychological or intellectual variables that most contributed to this condition.

Table 6 indicates that the stress score (29.5%) was the greatest predictor of academic burnout. This corresponds with the findings of Dyrbye et al. (2006), who indicated that persistent academic stress is a primary contributor to burnout among university students (GCU, 2024). Likewise, coping techniques assessed by the Brief COPE (26.4%) emerged as the second most significant component. Specifically, students employing maladaptive coping strategies—such as denial or substance use—exhibited elevated levels of burnout, aligning with the findings of Yang and Farn (2005), who highlighted the influence of coping behaviors on emotional tiredness (Wang et al., 2023).

Sleep duration (15.4%) and sleep quality (14.0%) were significant predictors. Insufficient or substandard sleep has been extensively associated with heightened emotional dysregulation and academic exhaustion (Lund et al., 2010; Becker et al., 2018; Bhat et al., 2023). These sleep-related issues may progressively diminish a student's capacity to recuperate from daily pressures, heightening susceptibility to burnout. Anxiety (5.9%) and sadness (5.6%), while less prevalent, nonetheless contributed significantly, reinforcing the notion that burnout frequently coincides with manifestations of emotional and psychological discomfort (Maslach & Leiter, 2016; Kebede et al., 2023).

Notably, year level (3.1%) contributed the least, suggesting that burnout is more related to individual psychological resilience than to academic advancement. Evidence indicates that students' mental health challenges endure over academic years when internal coping skills are insufficient (Sharma et al., 2022). The reduced incidence of burned-out students may also be associated with the gender composition of the respondents, predominantly male. Research conducted by Matud (2004) and Nolen-Hoeksema (2012) indicates that men typically employ more problem-focused coping strategies and have reduced tendencies toward rumination and emotional reactivity compared to women, potentially elucidating their diminished vulnerability to burnout (Lopez et al., 2023).

Table 6. Feature importance of predictors in the decision tree model

Predictor	Importance	Interpretation
Stress Score	29.5	Major Predictor
Brief Score	26.4	Major Predictor
Sleep Hours per Night	15.4	Strong Predictor
Sleep Quality	14.0	Moderate Predictor
Anxiety Score	5.9	Moderate Predictor
Depression Score	5.6	Moderate Predictor
Year Level	3.1	Weak Predictor

3.3.3 Model performance metrics

To evaluate the efficacy of the Decision Tree model in forecasting academic burnout, essential categorization metrics were analyzed. These encompass Recall (Sensitivity), F1 Score, and ROC-AUC. Each statistic offers insight into the model's efficacy in identifying at-risk students, managing predictive trade-offs, and differentiating between burnout and non-burnout categories.

Table 7 illustrates that the model attained a Recall of 51.6%, signifying that it accurately recognized approximately half of the students' experiencing burnout. This indicates intermediate sensitivity; however, it implies that certain at-risk pupils may remain unidentified, presenting a restriction when employed as the exclusive diagnostic instrument. The F1 Score of 59.3% indicates that the model achieves a reasonable equilibrium between accurately identifying burnout cases and minimizing false positives. Finally, the ROC-AUC score of 66.9% indicates that the model possesses a moderate capacity to differentiate between pupils experiencing burnout and those who are not. Hosmer et al. (2013) indicate that an AUC ranging from 60% to 70% signifies a model with acceptable yet insufficient discriminatory power. The results suggest that although the Decision Tree model is quite informative, its predictive accuracy might be enhanced by integrating more sophisticated or domain-specific variables, or by employing more advanced ensemble techniques.

Table 7. Model performance metrics

Metric	Value	Interpretation
Recall (Sensitivity)	51.6%	Model catches about half of true burnout students
F1 Score	59.3%	Model moderately balances correct predictions and false alarms.
ROC - AUC	66.9%	Model moderately distinguishes burnout vs. non-burnout students.

4 Conclusion

This research examined academic burnout in college students by employing psychometric and academic characteristics assessed via a Decision Tree algorithm. Although most respondents were not deemed to be experiencing burnout, nearly half (47.8%) were classed as at risk, with stress, coping mechanisms, and sleep patterns identified as the primary factors. The model exhibits modest accuracy in identifying burnout instances, evidenced by a recall of 51.6%, an F1 score of 59.3%, and a ROC-AUC of 66.9%. The comparatively lower burnout rate may be partially attributed to the sample's gender distribution, which was largely male—aligning with previous studies indicating that males generally display reduced emotional reactivity and rumination relative to females (Matud, 2004; Nolen-Hoeksema, 2012). In light of these findings, it is advisable for future studies to incorporate a more gender-balanced or female-majority sample and to investigate more sophisticated models, such as Random Forest, to improve predictive accuracy and obtain deeper insights into the determinants of academic burnout.

Consent

Permission to conduct the study was sought from the appropriate institutional authorities. The researcher ensured that participation in the study was entirely voluntary, and that informed consent was obtained from all student participants prior to data collection. Participants were assured that their responses would remain confidential and anonymous, with no names or personal identifiers included in any part of the analysis or reporting. The purpose, procedures, and scope of the study were clearly explained to all participants. They were informed of their right to refuse participation or withdraw from the study at any point without any academic or personal consequences. Data was collected only after securing consent, and the researcher-maintained transparency and honesty throughout the research process. Ethical guidelines for conducting research involving human participants were strictly followed.

Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

Competing Interests

Author has declared that they has no known competing financial interests or non-financial interests or personal relationships that could has appeared to influence the work reported in this paper.

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